

ObjectDNA

AI-Powered Circular Economy & Intelligent Reuse Platform

Global Innovation Hackathon 2026 — Project Submission

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1. Executive Summary

ObjectDNA is a circular-economy platform concept that helps users look beyond disposal. The intended experience is to scan an unwanted object, understand its visible condition and useful signals, compare practical next-life pathways and leave with a clear next action.

2. Problem Statement

Everyday objects are often discarded even when they still contain repairable parts, reusable materials or a possible second use. Users need a simple way to understand what can be repaired, reused, passed on or recovered.

3. Proposed Solution

- Object intake through an image.
- Object/material and condition signal identification.
- Component and remaining-value interpretation.
- Multiple next-life pathway generation.
- Pathway comparison using practical trade-offs.
- A clear recommended next action.

4. Prototype Features

- Responsive ObjectDNA landing and scanner experience.
- Image upload/dropzone with validation and local preview.
- Six deterministic demo objects.
- Four-phase analysis animation.
- Repair, reimagine, pass-on and recover comparison matrix.
- Interactive pathway selection and six-stage method selector.
- Keyboard accessibility, ARIA labels and reduced-motion support.
- Web manifest and service worker for an offline-first app shell.

5. Technology

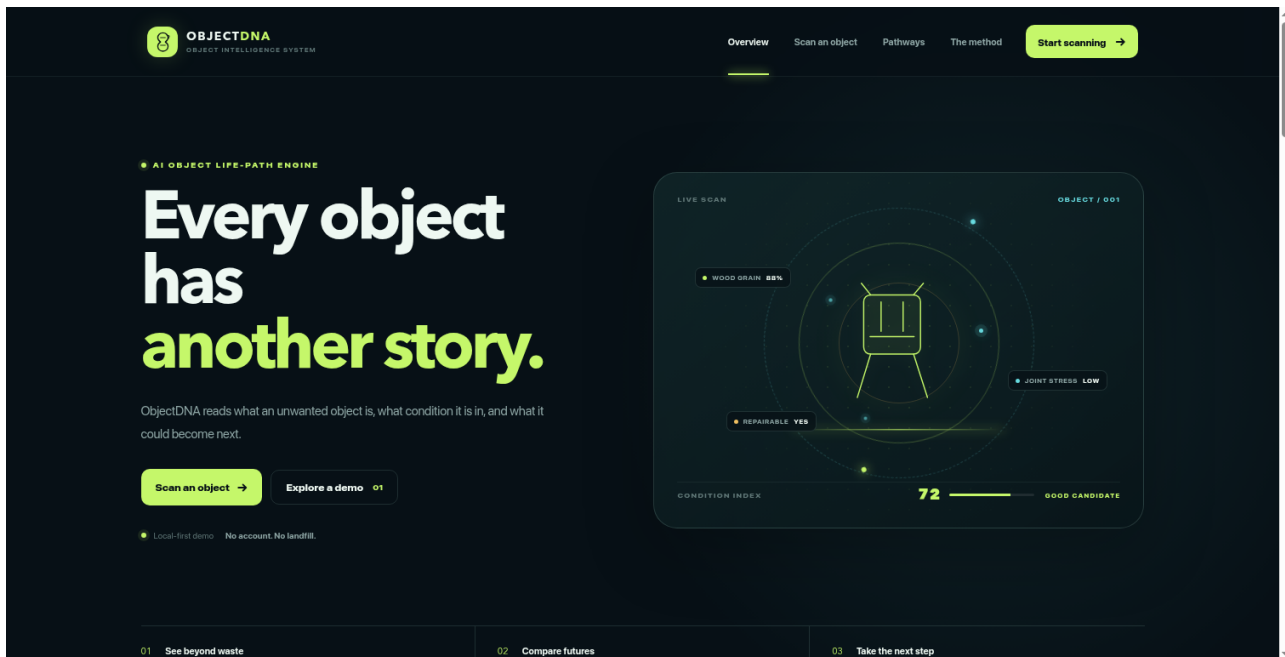
HTML5, CSS3, Vanilla JavaScript, Web Manifest, Service Worker, GitHub and Render static hosting.

6. AI / Backend Integration Note

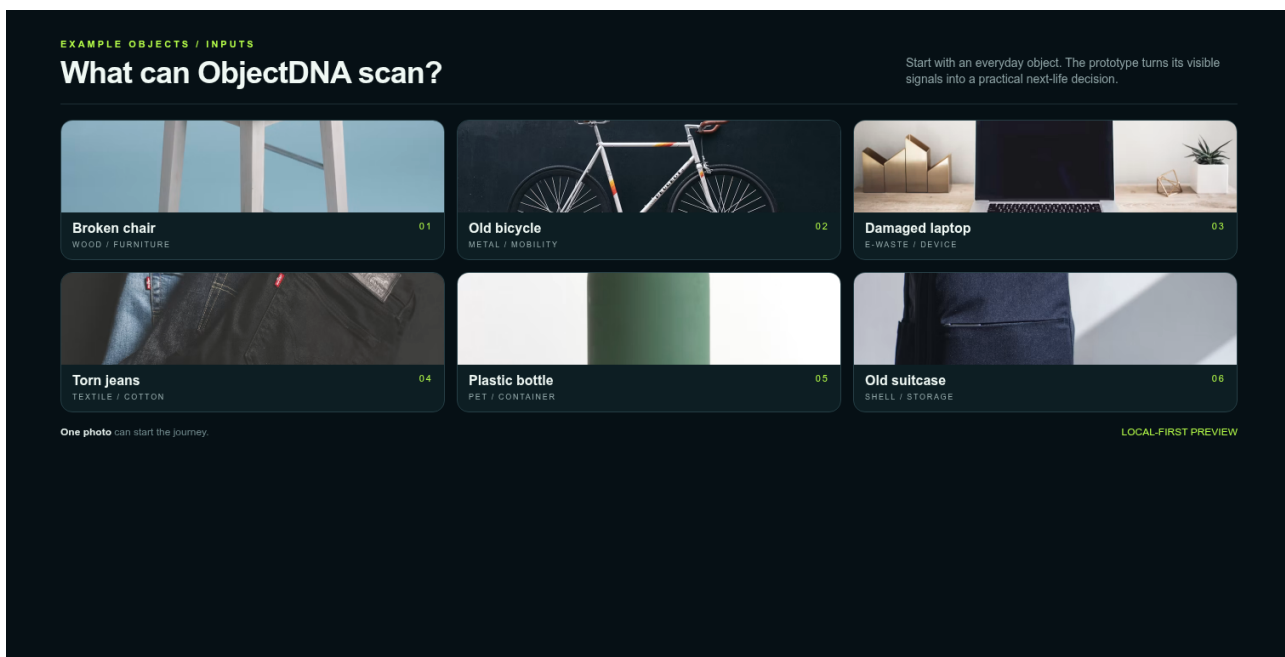
The submitted preview uses deterministic local demo data so the prototype can run without an external API key or backend. The app.js analysis boundary is the intended integration point for a secured vision/AI backend in a production implementation. This separation keeps the demo reliable while making the future model integration explicit.

7. Prototype Screens

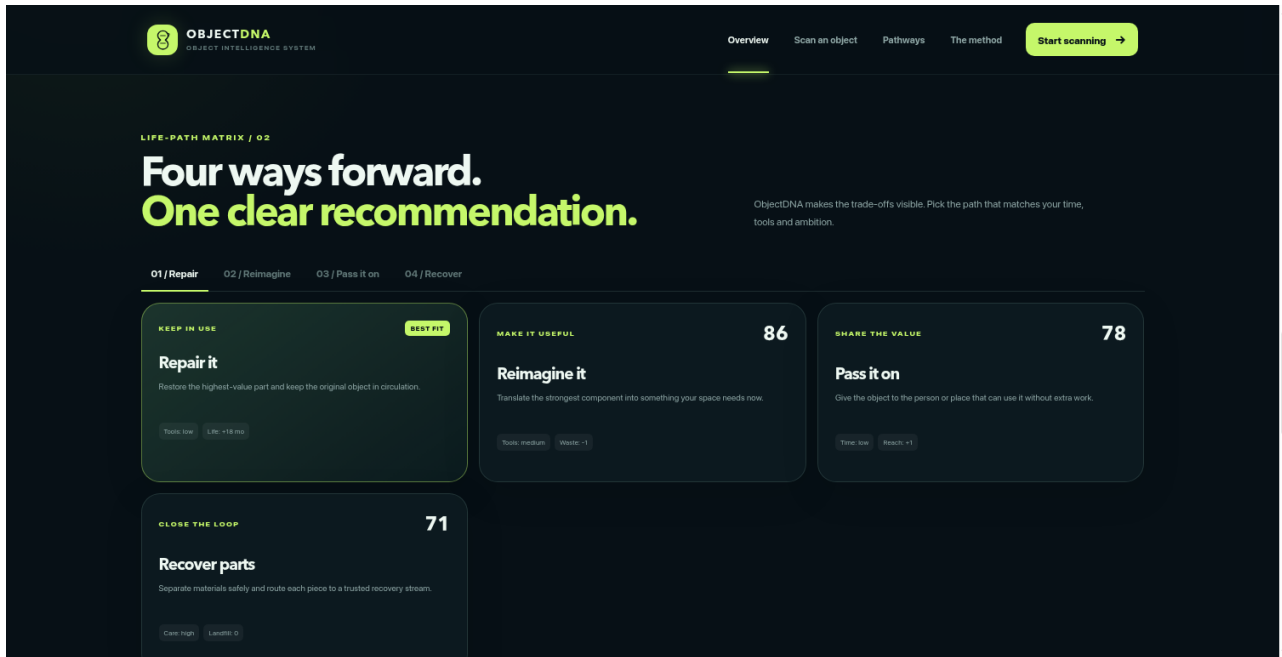
Homepage / ObjectDNA overview



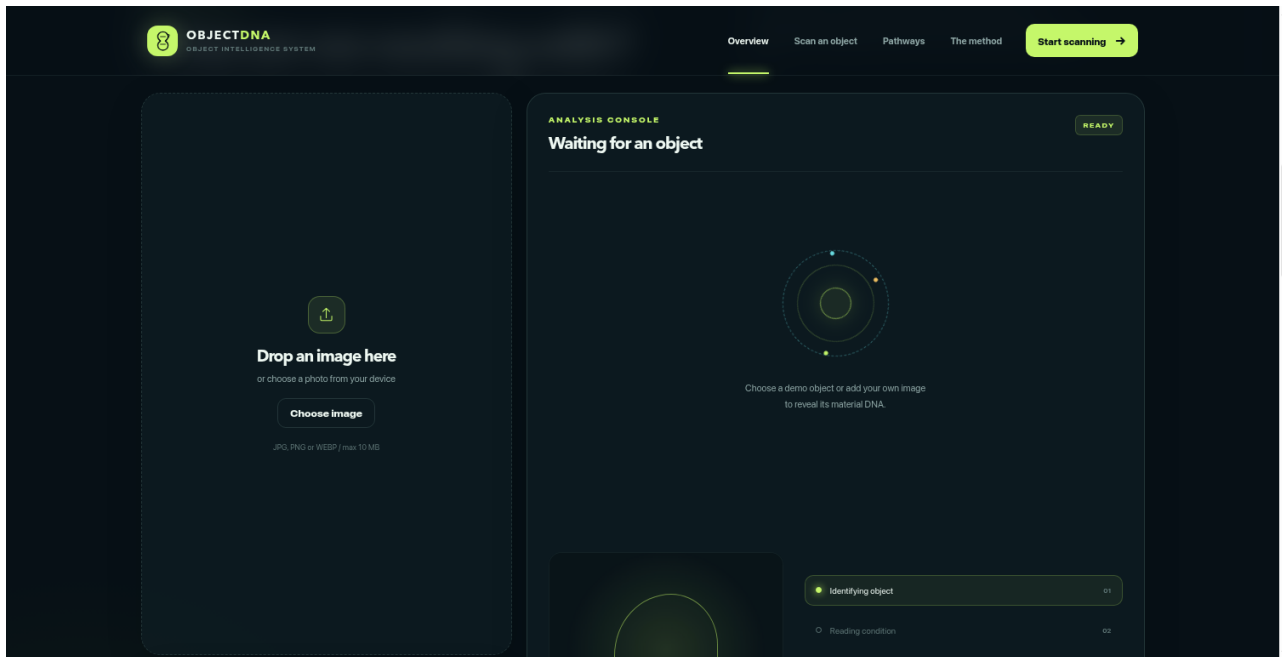
Example objects / supported inputs



Life-pathway comparison and recommendation



Object upload and analysis console



8. Circular-Economy Pathways

Pathway	Purpose
Repair it	Keep the original object in use by restoring useful parts.
Reimagine it	Repurpose strong remaining materials or components.
Pass it on	Transfer the object's remaining value to another user or place.
Recover parts	Separate useful materials/components for recovery.

9. Validation and Reliability

- Image type validation and 10 MB upload limit.
- FileReader and global runtime error feedback.
- Responsive desktop, tablet and mobile layouts.
- Keyboard-operable upload area and ARIA labels.
- Reduced-motion support and explicit empty states.
- No external CDN or package installation required.

10. Impact and Future Scope

ObjectDNA is designed to make practical circular actions easier to see and act on. Future versions can add a secured vision-language model, stronger condition/material detection, confidence scoring, local repair/donation/recycling discovery, user history, multilingual support, impact tracking and a reuse community marketplace.

11. Links

GitHub: <https://github.com/Aswini-ox/ObjectDNA-AI-Circular-Reuse-Platform>

Live prototype: <https://objectdna-ai-circular-reuse-platform.onrender.com>

Demo video: included in this ZIP under demo/ObjectDNA_Demo_Video.mp4